Intelligent Agents

CS44112: Artificial Intelligence

L-T-P-Cr: 3-0-2-4

Source: 1) NPTEL course on Artificial Intelligence
2) Artificial Intelligence: A Modern Approach, 3rd Edition, by Stuart Russell and Peter Norvig.

Objective

- Define an agent
- Define an Intelligent agent
- Define a Rational agent
- Explain Bounded rationality
- Discuss different types of environment- where agent operates
- Explain different agent architectures
- The various aspect of intelligent agents

Objective

On completion of this talk the student will be able to

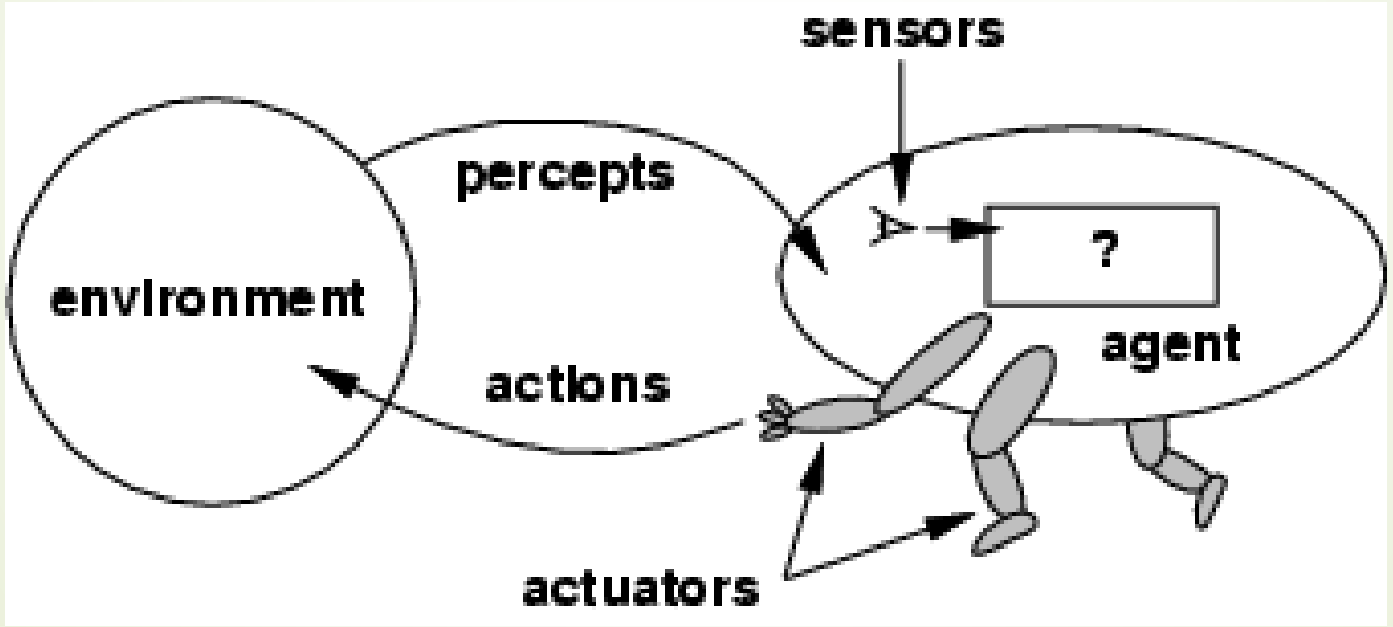
- Understand what an agent is and how an agent interacts with the environment. ***(environment is the important component for agent design and agent should be designed so that it can work properly in the environment)***
- Given a problem situation, the student should be able to
  - Identify the percepts available to the agent and
  - The actions that the agent can execute.
- Understand the performance measures used to evaluate an agent

Objective

- Be familiar with Diffrent agent architectures
  - Table based agent/ Simple reflex agents
  - State based agents
  - Goal-directed agents
  - Utility based agents
  - Learning agents
- Be able to analyze a problem situation and be able to
  - Identify the characteristics of the environment for a given problem situation
  - Recommended the architecture of desired agent

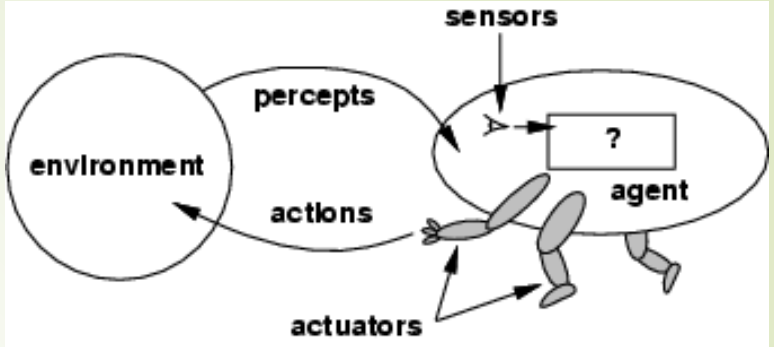
Agent and Environment

- **Definition: An agent perceives its environment via sensors and acts upon that environment through its actuators**



Agents interact with environments through sensors and actuators

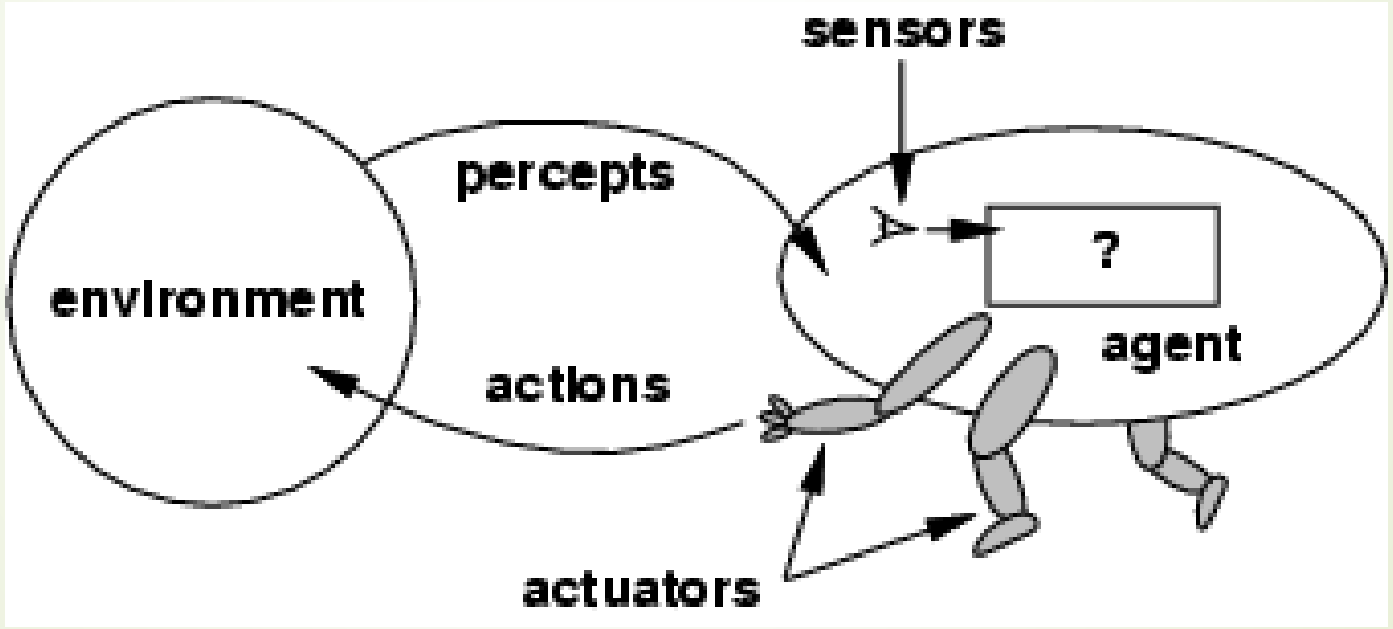
Agent and Environment



- Here agent operates in an environment
- Agent receives the percepts from environment
- And agent acts and its action change the environment
- Agent uses its various sensory organs, so depending on the sensors the agent has, **for example the agent may be able to see, hear , and accept the different inputs from environment**.
- Inside the agent, there is an **agent program** which decides on the basis of the current percepts, or the **percept sequence** that it has received till date to decide what should be a good action could be taken for current situation.
- So agent has **actuators and effectors** to take action, so these actions potentially change the environment and **agent can use its sensors to sense the changed environment.**

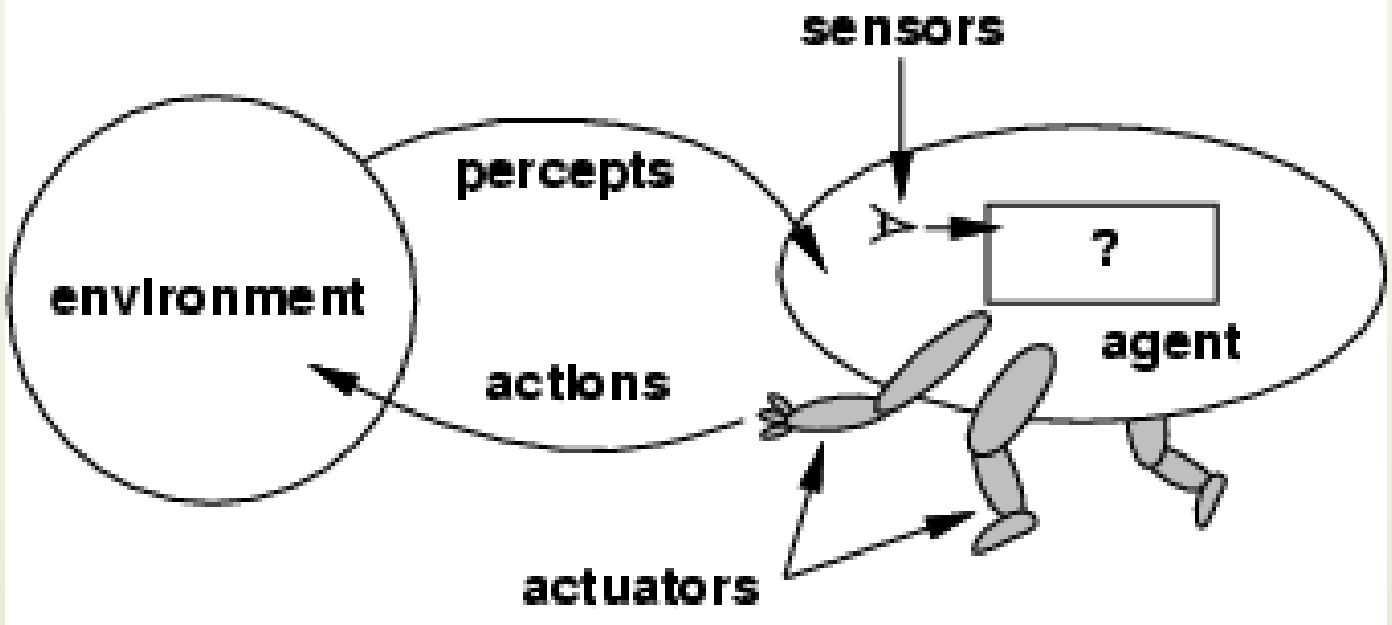
Agents

- **Have goals-** goals are the objectives that agents have to satisfy



Sensors and effectors

- An agent perceives its environment through sensors
  - The **complete set of inputs at a given time is called percepts**
  - The current percept, or a **sequence of percepts** (that the agent perceived so far) can influence the actions of an agent
- It can change the environment through effectors OR actuators
  - An operation involving an actuator is called an action
  - Actions can be grouped into **action sequences**



More on Agents

- We also have to **talk about performance measure to evaluate agents**
- Autonomous agent (In Artificial Intelligent agents, the autonomy is extremely important): decide **autonomously (without external control)** which action to take in the current situation to maximize progress towards its goals.

Performance

- Behavior and performance of IAs in terms of agent function
  - Perception history (sequence) to Action Mapping
  - Ideal mapping: specifies which actions an agent should to take at any point in time
- **Performance measure: a subjective measure to characterize how successful an agent is (e.g., speed, power usage, accuracy, money, etc.)**

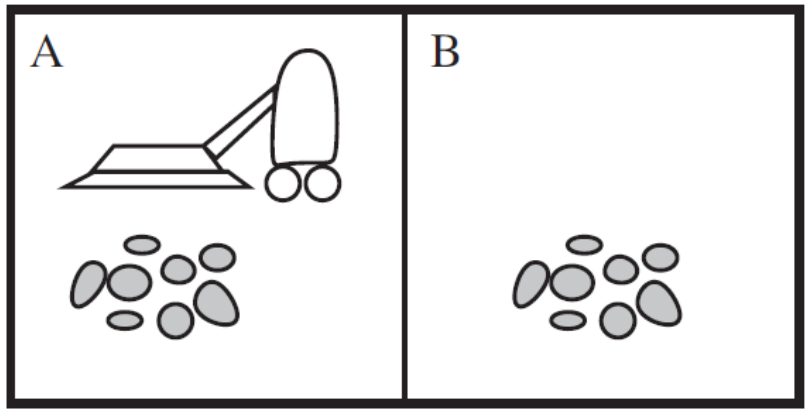
Example of Agents

- Humans
  - Eyes, ears, skin, taste buds, etc. for sensors
  - Hands, figures, legs, mouth for effectors
- Robots
  - Camera, infrared, bumper, etc. for sensors
  - Grippers, wheels , lights, speakers, etc. for actuators
- Software agent (softbot)
  - Functions as sensors
  - Functions as actuators

vacuum-cleaner agent

- This particular agent world has just two locations: squares A and B.
- The vacuum agent perceives which square it is in and whether there is **dirt** in the square. It can choose to move left, move right, suck up the dirt, or do nothing.
- One very simple **agent function** is the following: if the current square is dirty, then suck; otherwise, move to the other square.

vacuum-cleaner agent



A vacuum-cleaner world with just two locations.

Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Dirty]$	$Suck$
⋮	⋮
$[A, Clean], [A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Clean], [A, Dirty]$	$Suck$
⋮	⋮

Partial tabulation of a simple agent function for the vacuum-cleaner world

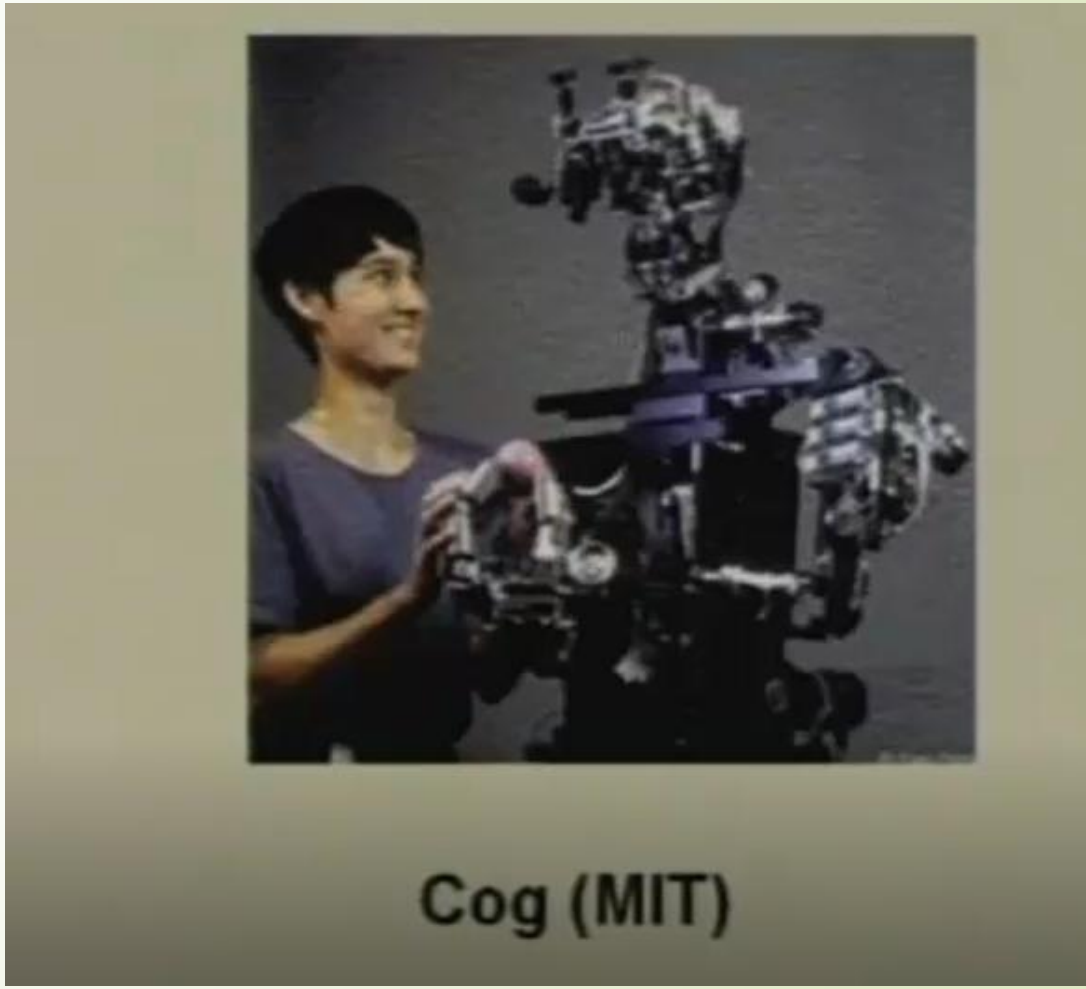
Types of Agents: Robots



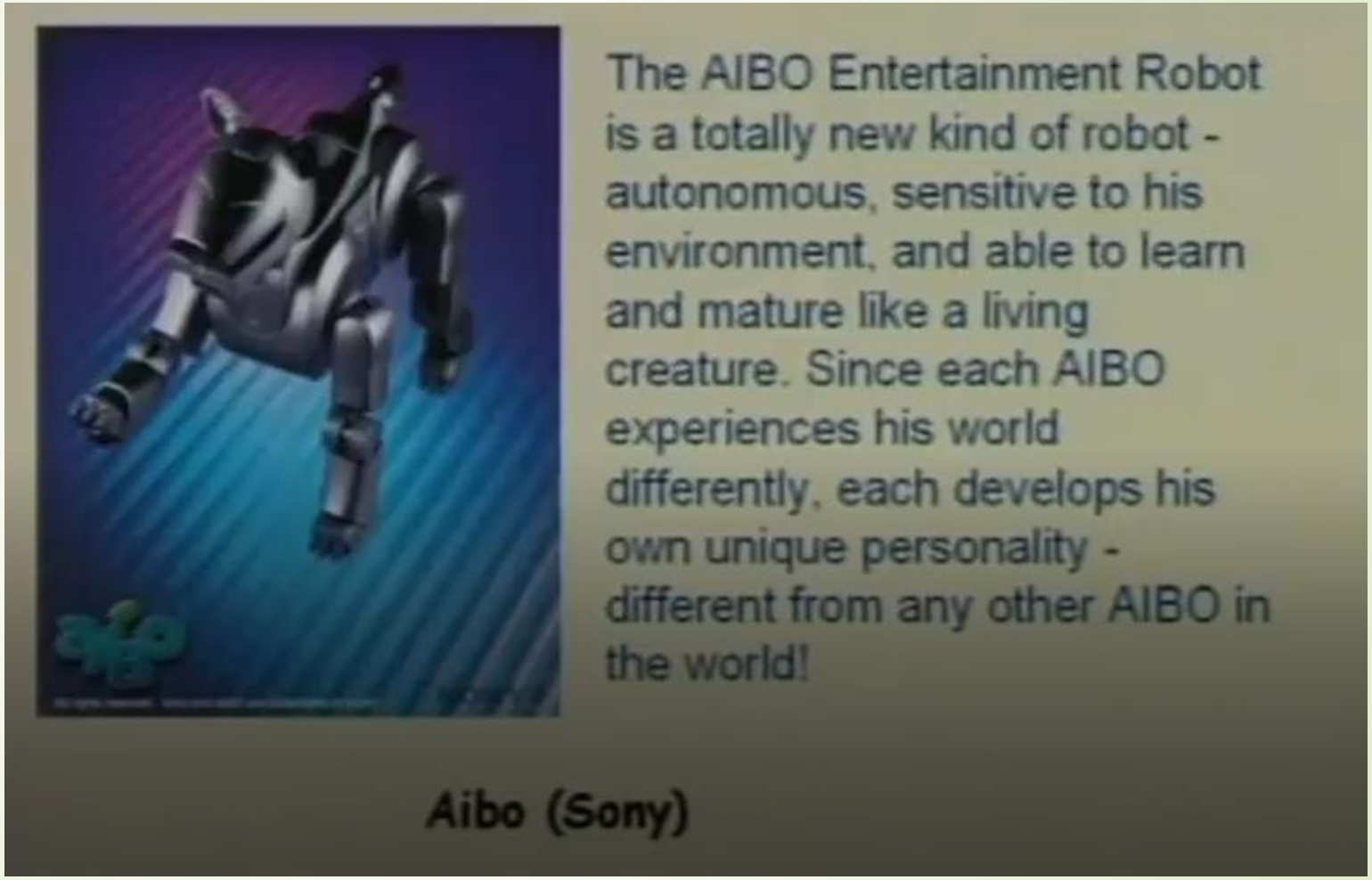
Types of Agents: Robots

- **Cog was developed by Rodney Brooks group at MIT**
- **Homepage of Cog:**

- **They wanted to build humanized robot, i.e. robot has the intelligence like human being.**



Types of Agents: Robots



Other types of Agents

- Softbots
  - Askjeeves.com // search engine launched in 1997
- Expert Systems (medical expert systems) //providing expert advice
- Autonomous spacecraft (i.e. Marsh robot) // that can sense their environment, make decisions, and execute tasks without direct human intervention
- Intelligent buildings (to decide lighting, air conditioning etc..)

Specifying the task environment: PEAS (Performance, Environment, Actuators, Sensors) description

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Fig. PEAS description of the task environment for an automated taxi.

Specifying the task environment: PEAS (Performance, Environment, Actuators, Sensors) description

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Intelligent Agents

- Fundamental faculties of intelligence
  - Acting
  - Sensing
  - Understanding, reasoning, learning **(it is important to operate in unknown environment thats why learning is required),** learning is the prerequisite for agent to be autonomous. Autonomous agent which can adjust to change in the environment and must have some learning component.
- **Intelligent agent must be rational**

Rational Agent

- AI is about building rational agents.
- An agent is something that perceives and act.
- A rational agent always does the right thing. Right thing?
  - What are the functionalities (goals)?
  - What are the components?
  - How do we build them?

Rationality

- Perfect Rationality
  - Assumes that the rational agent knows all and will take the action that maximizes her utility.
  - Human beings do not satisfy this definition of rationality. However perfect rationality is something that is not within our scope. Many times we cannot reason in a given time frame.
- Bounded Rationality Herbert Simon, 1972 (of CMU)
  - Because of the limitations of the human mind, humans must use approximate methods to handle many tasks.
  - The bounded rationality does not have the aim to maximize the utility. So it is import for agent to take best action by given limited resources.

Rationality

- **Rational Action: The action that maximizes the expected value of the performance measure given the percept sequence.**
  - When agents takes action then action may not always have the same outcome.
  - Sometimes the outcome of the action may be good, may be bad. So when we talk about the rationality then we must look at the expected value of the performance measure NOT always the absolute value of the performance measure.
  - Also we should evaluate an agent not on the basis of how it should behave in perfect case but how it can behave based on what it can sense.
- Rational= Best?
  - Yes, to the best of its knowledge
- Rational= Optimal?
  - Yes, to the best of its abilities
  - And its resources/ constraints

Agent Environment

- Environment in which agents operate can be defined in different ways.

Environment characteristics?

Environment based on Observability

- Fully observable
  - All of the environment relevant to the action being considered is observable
  - Such environments are convenient, since the agent is freed from the task of **keeping track of the changes in the environment**.
- Partially observable
  - The relevant features of the environment are only partially observable.
- Example:
  - Fully obs: Chess (agent has complete knowledge of board, so everything about the environment is accessible to the agent)
  - Partially obs: Poker (so agent cannot see the hand of the opponent so environment in this case is only partially observable to the agent)

Environment types

- **Deterministic:** If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic. Image analysis
- **Stochastic**: If uncertainty occurs then the environment is stochastic. Note that a deterministic yet partially observable environment will *appear* to be stochastic to the agent. Ludo
- **Strategic**: environment state wholly determined by the preceding state and the actions of *multiple agents is called strategic*. *Chess*
- *The environment is deterministic OR stochastic depends on how you look the environment, if you cannot observe the environment fully then environment may appear stochastic to you whereas it is actually the deterministic.*

Environment: Episodicity

- Episodic/Sequential
  - An **episodic** environment means that subsequent episodes do not depend on what actions occurred in previous episodes.- sometimes certain task can be divided into different phases/episodes, in such episodic environment, one episode may or may not influence the subsequence episodes.
  - In an **sequential** environment, the agent engages in a series of connected episodes. –So subsequent episodes are dependent on previous episode.
    - Interactive Tutor

Environment: Dynamism

- **Static Environment:** does not change from one state to the next while the agent is considering its course of action.
- **Dynamic Environment:** Changes over time independent of the actions of the agent.

Environment: Continuity

- **Discrete/Continuous**
  - If the number of distinct percepts and the actions is limited, the environment is **discrete**, otherwise is continuous.

Complex Environments

- Complexity of the environment includes
  - **Knowledge rich:** enormous amount of information that the environment contains and
  - **Input rich:** the enormous amount of input the environment can send to an agent.
- The agent must have a way of managing this complexity. Often such considerations lead to the development of
  - Sensing strategies and – what to sense, what is more relevant
  - Attentional mechanisms- what it should give the attention too
- So that the agent may more readily focus its efforts in such rich environments.

Examples of task environments and their characteristics

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Agent architectures
Table based agent/ Simple reflex agents

- Table based is the very simple agent- here mapping between the percepts and actions is stored in the form of table (left side we have percepts and right side we have actions)
- Information comes from **sensors- percepts**
- Triggers **actions** through the **effectors**

When action is only dependent on current percepts, then it is easy to develop table based agent. But it is infeasible when the correct or best action depends on the past.

Table based agent/ Simple reflex agents

- These agents select actions on the basis of the *current percept,* ***ignoring the rest of the percept history*****.**
- ***For example****, the vacuum agent* whose agent function can tabulated is a simple reflex agent, because its decision is based only on the current location and on whether that location contains dirt.

Goal based Agent

- Goal-based AI agents represent a sophisticated approach in artificial intelligence (AI), where agents are programmed to achieve specific objectives.
- These agents are designed to plan, execute, and adjust their actions dynamically to meet predefined goals, For Example: A **self-driving car** using planning and reasoning to navigate to a set destination **(the goal)** by evaluating routes, adapting to traffic, and avoiding obstacles.
- This approach is particularly useful in complex environments where flexibility and adaptability are crucial.

Utility based Agent

- Goals alone are not enough to generate high-quality behavior in most environments.
- For example, many action sequences will get the taxi to its destination (thereby achieving the goal) but some are quicker, safer, more reliable, or cheaper than others.
- Goals just provide a crude binary distinction between “happy” and “unhappy” states.

Learning Agent

- Learning allows an agent to operate in initially unknown environment.
- The learning element modifies the performance element.
- Learning is required for true autonomy.

Summary

- An **agent** perceives and acts in an environment, has an architecture, and is implemented by an agent program.,
- An **ideal agent** always chooses the action which maximizes its expected performance, given its percept sequences so far.
- An **autonomous agent** uses its own experience rather than built-in knowledge of the environment by the designer.

Summary

- An **agent program** maps from percepts to action and updates its internal state.
  - Goal-based agents act in order to achieve their goal (s).
  - Utility-based agents maximize their own utility function.
- **Representing knowledge** is important for successful agent design.
- The most challenging environments are **partially observable**, **stochastic**, **sequential**, **dynamic**, and **continuous, and contain multi intelligent agents**.

Questions

- Define an agent.
- What is a rational agent?
- What is bounded rationality?
- What is an autonomous agent?
- Describe the salient features of an agent.

Questions

- Find out about the Mars rover.
  - What are the percepts for this agent?
  - Characterize the operating environment.
  - What are the actions the agent can take?
  - How can one evaluate the performance of the agent?
  - What sort of agent architecture do you think is most suitable for this agent?
- Answer the same questions as above for an Internet shopping agent.

End